



Junwoo Hong

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I am an undergraduate at POSTECH, currently a research intern at the Data Systems Lab under Prof. Wook-Shin Han.

Most of what I build starts from a personal pain point — something tedious I had to do by hand, or a workflow that felt like it should already be automated. Assignment deadlines that never made it into my calendar, lecture recordings I was never going to revisit, school mail I kept sorting manually. I like the gap between “this is annoying” and “this is fixed.”

Lately the same instinct has pointed at language-model agents: they do a great deal in the middle of a task, and almost none of it can be checked. That is what I work on now.

RESEARCH INTERESTS

- Memory representations for LLM agents — what an agent records between turns, and in what form
- Verifiable process reward: intermediate signals that can be scored by an external checker for RL
- Multi-hop retrieval, and attribution of intermediate conclusions to evidence
- Evaluation methodology for agent benchmarks — arm symmetry, harness artifacts, claim boundaries

EDUCATION

B.S. in Computer Science and Engineering, Pohang University of Science and Technology 2025.03 – present
(POSTECH)

Coursework in data structures, algorithms, systems programming, and machine learning.

EXPERIENCE

Undergraduate Research Intern, Data Systems Lab (DSLAb), POSTECH 2026.06 – present

Advised by Prof. Wook-Shin Han. Working on PROVE (below).

RESEARCH

PROVE: Provenance for Retrieval via Observable, Verified Evidence 2026.06 – present

Multi-hop QA agents hold their working memory as a free-form blob, so nothing they conclude mid-search can be checked against the evidence behind it. PROVE puts a verifiable scratchpad next to free-form reasoning — think in prose, answer through the scratchpad — and asks whether that yields intermediate reward usable for RL without costing accuracy. Currently moving from evaluation to RL training.

Semantic and Graph Context for Personal Knowledge Bases 2026 – present

Independent work on serving a personal note graph to language-model agents: local embedding search combined with link structure, exposed through the Model Context Protocol under an explicit write guard. Released as [obsidian-everywhere](#).

SOFTWARE

- 1. **obsidian-everywhere** — Graph and local semantic context for Obsidian, with guarded remote MCP access and edits from any MCP client. `TypeScript`
- 2. **athens** — Pairwise music rating application; ratings are elicited by comparison rather than absolute score. Flutter, Supabase, Next.js. `Dart`
- 3. **justpasteyourplaylist** — Converts a Spotify or Apple Music playlist link into a shareable YouTube playlist without API keys. `TypeScript`
- 4. **altToNotes** — Aligns lecture voice recordings with slide decks and generates per-page explanatory notes with an LLM. `Python`
- 5. **AutoMail** — macOS client that classifies university IMAP mail into four categories on device. `Swift`
- 6. **plms2Calendar** — Synchronises POSTECH PLMS coursework deadlines to Google Calendar. `Python`

Full repository list →

TECHNICAL BACKGROUND

Languages	Python, TypeScript/JavaScript, Swift, C/C++, Dart
Frameworks	React, Next.js, Flutter, SwiftUI
Infrastructure	Supabase, Docker, Git, Unix
